Adam Ormondroyd

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Education

University of Cambridge: Natural Sciences M.Sci. (Physical) *Sep. 2017 – July 2021*

Part IA	Computer Science (including algorithms and numerical methods), Materials Science, Mathematics, and Physics.	1st (75.4%)
Part IB	Double Physics and Materials Science.	1st (76.7%)
Part II	Physics: my courses included Advanced Quantum, Astrophysical Fluid Dynamics, Particle and Nuclear Physics, and Relativity.	1st (79.2%)
Part III	Physics: I achieved the highest mark in the cohort in both the Particle Physics and Relativistic Astrophysics and Cosmology major exams. My minor exams included Gauge Field Theory and Advanced Statistical Mechanics.	1st (80.0%)

Beckfoot School, Bingley *Sep. 2010 – June 2017*

A-level	Computer Science: A*, Maths: A*, Further Maths: A*, Physics: A*.
AS-level	Chemistry: A.
GCSE	One Aˆ, eight A*s, two As and one B, including English and maths.

Employment

Data Scientist, PolyChord Ltd *Oct. 2025 – present*

My work centres on MIDAS (Maximum Information Data Acquisition Strategies), which combines the OODA loop (observe, orient, decide, act) with Bayesian experimental design, using inference tools from cosmology as the glue. I am first author of the technical reports for our project funded by the Defence Science and Technology Laboratory (Dstl). My feedback as an early adopter informed the design of the BlackJAX nested sampler, to which I continue to contribute upstream. PolyChord's successful Innovate UK Frontier AI Discovery bid builds on the MIDAS drone simulation I developed during my internship; I also produced technical work for the bid and reviewed the proposal text.

CDT internship at PolyChord Ltd: MIDAS *Oct. 2023 – March 2024*

I was the primary developer of a simulated capture-the-flag game in which MIDAS directs drones among buildings, on a Defence and Security Accelerator (DASA) project; I also improved MIDAS's multiprocessing, addressed assumptions built into its prior, and began an alternative approach using simulation-based inference.

Keighley & Worth Valley Railway *Summers 2017 and 2018*

My roles included crossing keeper (operating signals), station foreman, booking clerk and on-train ticket inspector, and I was occasionally assigned track walks to inspect and make minor repairs to the line. The staff also asked me to reorganise their database in quiet moments between trains.

Software and Programming

During my undergraduate degree, I used Java and functional programming in my computer science courses, and Python and Matlab for scientific computing. Through my PhD research, I have worked primarily in Python and Fortran, with some C++. Recently, I have been using GPU acceleration with JAX. Beyond my research, I have written C and rust, and got Fortran talking to rust through C++ to see if I could. I make habitual use of git for version control of all my projects. I have made contributions to open-source scientific software, including the widely-used Python libraries pandas and matplotlib, via github. I took the lead in keeping my research group's various packages up to date with the latest Python dependencies, particularly the sample post-processing package anesthetic, which integrates tightly with pandas. I also made improvements to the nested sampling tool PolyChord, which is used by many research groups worldwide.

I believe that less code is more. My package janesthetic reimplements the key calculations of anesthetic in about 140 lines of JAX, with tests confirming that it matches anesthetic's results. I also wrote jesi, a minimal JAX library of cosmological likelihoods, which I used for my paper "Dynamic or Systematic?". My flexknot library, from my PhD, has since been used in a colleague's research.

Research

PhD: "Bayesian methods in a flexible universe: sampling, tension and dark energy" *Oct. 2021 – Sep. 2025* *Awarded May 2026*

Supervisors: M.P. Hobson, W.J. Handley, A.N. Lasenby

In cosmology, there are many cases where it is useful to reconstruct one–dimensional functions, such as the primordial matter power spectrum or the evolution of the dark energy equation of state. I investigated a free-form approach: a linear spline between a variable number of nodes, using Bayesian statistics to make comparisons between these and other models. A wide range of cosmological datasets were used, including the cosmic microwave background, baryon acoustic oscillations, and Type Ia supernovae. In addition to computing the Bayesian evidence for model comparison, I also used statistics to quantify the possibility of tension between these datasets, and how this tension evolves with model choice.

I also investigated the effects of cluster detection for mode detection and evidence calculation in nested sampling of multi-modal posterior probability distributions.

I wrote a first-author paper on the DESI DR2 results within two days of their release; it was later published in MNRAS. I have presented my work at EuCAIFCon I and II (flash talks and posters) and GR/Amaldi (talk).

Ozone damage in broadleaf saplings from reflectance spectra *Nov. – Dec. 2023*

Tropospheric ozone is a pollutant which can damage vegetation. Hyperspectral reflectance spectra of leaves can be used to identify signatures of different stressors. I became involved in this project through a collaboration with Dr. Anna Jones, who had mentioned that the standard analysis pipeline would take many hours to run. I got it down to under a second!

Part III Research Project "Controlling the Beast" *Oct. 2020 – May 2021*

The Magdalena Ridge Observatory Interferometer has a problem with its prototype hardware: the mirrors used to steer light beams from the telescopes to the combining facility are thermally unstable, causing the light beams to wander in angle unacceptably. My project used temperature measurements of the components to understand the dominant causes of this motion, and developed a predictive model to be used to control a corrective system. I also identified issues with the experimental setup which will be corrected in future tests. I learned to use advanced statistical techniques including principal component analysis and singular value decomposition, and machine learning using the Keras API for TensorFlow.

Blundell Lab, University of Cambridge *June 2020 – Sep. 2020*

I remotely investigated a dataset of ultra-deep genome sequences from the Avon Longitudinal Study of Parents and Children (ALSPAC) to create a model for the frequency of sequencing errors in blood samples. I conclusively showed that the position-specific Poisson noise model usually used to identify low-frequency variants is unsuitable. I improved my data analysis skills, in particular in dealing with using files much larger than my laptop's RAM, and gained insight into working in a different field.

Gravitational Physics Group, Cardiff University *July 2019 – Aug. 2019*

I designed and simulated an interferometer using the Finesse software via Python to investigate the holographic principle. Previous setups had 40m arms; the challenge was to fit in a cleanroom only 5m wide while increasing the sensitivity! I also attended meetings discussing the most recent LIGO signals; I enjoyed seeing the collaboration and discussion determining if they were gravitational waves. I shared my own work during these meetings, and the fresh perspectives of the other researchers provided a different approach key to solving my project.

Publications

- A.N. Ormondroyd, W.J. Handley, M.P. Hobson, A.N. Lasenby, D. Yallup, "Dynamic or systematic? Bayesian model selection between dark energy and supernova biases", *MNRAS* (2026).
- A.N. Ormondroyd, W.J. Handley, M.P. Hobson, A.N. Lasenby, "Comparison of dynamical dark energy with ΛCDM in light of DESI DR2", *MNRAS* (2026).
- S. Legner, W.J. Handley, W. Barker, A.N. Ormondroyd, "Alleviating the Hubble tension with Torsion Condensation (TorC)", *JCAP* (2026).
- A.N. Ormondroyd, W.J. Handley, M.P. Hobson, A.N. Lasenby, "Non-parametric reconstructions of dynamical dark energy via flexknots", *MNRAS* (2025).

- A.L. Jones, A.N. Ormondroyd, F. Hayes, E.S. Jeffers, "Reflections of stress: Ozone damage in broadleaf saplings can be identified from hyperspectral leaf reflectance", *Environmental Pollution* (2024).
- A.N. Ormondroyd, W.J. Handley, M.P. Hobson, A.N. Lasenby, "Tilting the scales: weighing prior dependency and global tensions of CMB lensing", preprint, arXiv:2310.08490 (2023).

Teaching

- During my PhD, I supervised Part IA NST Mathematics for both Queens' and Pembroke colleges: 40 students over four years, 365 hours of teaching.
- Through 6th form, I assisted teaching year 7 music classes and after-school small-group sessions with year 10 and 11 students requiring extra help.

Extracurricular

- Treasurer of the Cambridge University Automobile Club for four years and Championship Secretary for two, organising a record number of race events in 2024–25.
- Twice club champion (2023–24 and 2025–26), with 13 wins, 5 second places and 9 third places. I contributed to four consecutive Varsity wins, winning the race twice and finishing second and third in the others, and briefly held the lap record at Red Lodge. I also placed third in British Universities Karting Championship (BUKC) qualifying races in 2022 and 2023, both helping to qualify the team for the main championship.
- Coxed Queens' College M3 for a year and M1 the following Michaelmas term.